

Mitchell Waghorn

Electrical Engineering & Computer Science Student

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PROFILE

Electrical Engineering and Computer Science student at the University of New South Wales with a strong interest in systems programming, embedded systems, low-level software, and software engineering. Experienced in Rust, C, Java, Python, and modern software development through university coursework and personal projects. Seeking opportunities to apply both hardware and software engineering skills in industry.

EMPLOYMENT HISTORY

Application Technician

Jul 2023 — Present

Self-Employed — Sydney

- Independently manage all aspects of vinyl advertising installation, from client acquisition to project execution.
- Install advertising for the Sydney Light Rail and contribute to anti-graffiti campaigns for the City of Sydney.
- Personally handle invoicing, scheduling, and customer relations to ensure efficient project completion.
- Maintain and apply self-adhesive vinyl to vehicles, shopfronts, and homes while delivering high-quality results.
- Adapt installation solutions to individual client requirements while complying with relevant state and federal regulations.

Application Technician

Aug 2020 — Jun 2023

Blindlight — Sydney

- Assisted during the growth of a small business by installing products, invoicing, and managing small to medium-scale jobs.
- Worked with large corporate clients to ensure advertising installations met specifications and project deadlines.

Long-Life Team Member

Apr 2015 — Aug 2020

Woolworths — Carlingford

- Worked full time in one of Sydney's highest foot-traffic stores, maintaining stock availability and inventory accuracy.
- Delegated tasks to team members during team meetings and supported daily store operations.
- Assisted other stores with stocktake organisation and long-term inventory management.

EDUCATION

Bachelor of Electrical Engineering & Computer Science

Jun 2023 — Present

University of New South Wales — Sydney

- Current student in the Electrical Engineering and Computer Science double degree program.
- 84.95 WAM
- COMP3121: Algorithm Design and Analysis – 86
- COMP6991: Solving Modern Programming problems with Rust – 94
- COMP2511: O-O Design and Programming - 95

TECHNICAL SKILLS

Programming	Rust, C, Java, Python, JavaScript, Claude Code
Web	React, HTML, CSS
Engineering	Embedded systems, circuits, electronics, MATLAB
Professional	Teamwork, Agile development, Client management, Project coordination

PROJECTS

Riskit – Online Game Website

TypeScript, React

- Developed a web-based game platform featuring simulated gambling-style games, including minesweeper, Higher or Lower, and Blackjack.
- Implemented a competitive leaderboard system in which users compete for rankings using virtual currency rather than real-money wagering.
- Built the frontend using TypeScript and React, focusing on reusable components, game logic, and interactive user interfaces.

ARM7TDMI-S Embedded Home automation System

C, ARM Assembly, Embedded Systems

- Developed an embedded home automation system using C and ARM Assembly on an ARM7TDM-S platform.
- Integrated external peripherals including light and temperature sensors and a capacitive touchscreen LCD for environmental monitoring and user interaction.
- Implemented automated control of simulated household systems including window blinds, a doorbell, coffee machine and climate control.
- Performed hardware integration, testing, and validation using an Embedded Artists ARM Development board.

Oscillite – Mixed-Signal Synthesizer (In Development)

C++, Teensy 4.1, DSP, Embedded Systems

- Developing a mixed-signal synthesizer on the Teensy 4.1, combining embedded software, digital signal processing, and analogue electronics.
- Designed the software around real-time and resource constraints, using fixed point arithmetic to reduce computational and memory overhead compared with floating -point processing.
- Developed a custom Q8 fixed-point arithmetic library for efficient numerical processing on the embedded platform.
- Implemented core synthesizer components including ADSR envelope generator and digital waveform generator.
- Deigning the project as an integrated hardware/software system, applying concepts from both electrical engineering and computer science.